CHAPTER 27 Martingales and Measures

Up to now interest rates have been assumed to be constant when valuing options. In this chapter, this assumption is relaxed in preparation for valuing interest rate derivatives in Chapters 28 to 32.

The risk-neutral valuation principle states that a derivative can be valued by (a) calculating the expected payoff on the assumption that the expected return from the underlying asset equals the risk-free interest rate and (b) discounting the expected payoff at the risk-free interest rate. When interest rates are constant, risk-neutral valuation provides a well-defined and unambiguous valuation tool. When interest rates are stochastic, it is less clear-cut. What does it mean to assume that the expected return on the underlying asset equals to the risk-free rate? Does it mean (a) that each day the expected return is the one-day risk-free rate, or (b) that each year the expected return is the 1-year risk-free rate, or (c) that over a 5-year period the expected return is the 5-year rate at the beginning of the period? What does it mean to discount expected payoffs at the risk-free rate? Can we, for example, discount an expected payoff realized in year 5 at today's 5-year risk-free rate?

In this chapter we explain the theoretical underpinnings of risk-neutral valuation when interest rates are stochastic and show that there are many different risk-neutral worlds that can be assumed in any given situation. We first define a parameter known as the *market price of risk* and show that the excess return over the risk-free interest rate earned by any derivative in a short period of time is linearly related to the market prices of risk of the underlying stochastic variables. What we will refer to as the *traditional risk-neutral world* assumes that all market prices of risk are zero, but we will find that other assumptions about the market price of risk are useful in some situations.

Martingales and *measures* are critical to a full understanding of risk neutral valuation. A martingale is a zero-drift stochastic process. A measure is the unit in which we value security prices. A key result in this chapter will be the *equivalent martingale measure result*. This states that if we use the price of a traded security as the unit of measurement then there is some market price of risk for which all security prices follow martingales.

This chapter illustrates the power of the equivalent martingale measure result by using it to extend Black's model (see Section 17.8) to the situation where interest rates are stochastic and to value options to exchange one asset for another. Chapter 28 uses the result to understand the standard market models for valuing interest rate derivatives, Chapter 29 uses it to value some nonstandard derivatives, and Chapter 31 uses it to develop the LIBOR market model.

27.1 THE MARKET PRICE OF RISK

We start by considering the properties of derivatives dependent on the value of a single variable θ. Assume that the process followed by θ is

$$\frac{d\theta}{\theta} = m\,dt + s\,dz \tag{27.1}$$

where dz is a Wiener process. The parameters m and s are the expected growth rate in θ and the volatility of θ, respectively. We assume that they depend only on θ and time t. The variable θ need not be the price of an investment asset. It could be something as far removed from financial markets as the temperature in the center of New Orleans.

Suppose that f_1 and f_2 are the prices of two derivatives dependent only on θ and t. These can be options or other instruments that provide a payoff equal to some function of θ at some future time. Assume that during the time period under consideration f_1 and f_2 provide no income.[1]

Suppose that the processes followed by f_1 and f_2 are

$$\frac{df_1}{f_1} = \mu_1\,dt + \sigma_1\,dz$$

and

$$\frac{df_2}{f_2} = \mu_2\,dt + \sigma_2\,dz$$

where μ_1, μ_2, σ_1, and σ_2 are functions of θ and t. The "dz" in these processes must be the same dz as in equation (27.1) because it is the only source of the uncertainty in the prices of f_1 and f_2.

The prices f_1 and f_2 can be related using an analysis similar to the Black–Scholes analysis described in Section 14.6. The discrete versions of the processes for f_1 and f_2 are

$$\Delta f_1 = \mu_1 f_1\,\Delta t + \sigma_1 f_1\,\Delta z \tag{27.2}$$

$$\Delta f_2 = \mu_2 f_2\,\Delta t + \sigma_2 f_2\,\Delta z \tag{27.3}$$

We can eliminate the Δz by forming an instantaneously riskless portfolio consisting of $\sigma_2 f_2$ of the first derivative and $-\sigma_1 f_1$ of the second derivative. If Π is the value of the portfolio, then

$$\Pi = (\sigma_2 f_2) f_1 - (\sigma_1 f_1) f_2 \tag{27.4}$$

and

$$\Delta\Pi = \sigma_2 f_2\,\Delta f_1 - \sigma_1 f_1\,\Delta f_2$$

Substituting from equations (27.2) and (27.3), this becomes

$$\Delta\Pi = (\mu_1\sigma_2 f_1 f_2 - \mu_2\sigma_1 f_1 f_2)\Delta t \tag{27.5}$$

[1] The analysis can be extended to derivatives that provide income (see Problem 27.7).

Because the portfolio is instantaneously riskless, it must earn the risk-free rate. Hence,

$$\Delta\Pi = r\Pi\,\Delta t$$

Substituting into this equation from equations (27.4) and (27.5) gives

$$\mu_1\sigma_2 - \mu_2\sigma_1 = r\sigma_2 - r\sigma_1$$

or

$$\frac{\mu_1 - r}{\sigma_1} = \frac{\mu_2 - r}{\sigma_2} \tag{27.6}$$

Note that the left-hand side of equation (27.6) depends only on the parameters of the process followed by f_1 and the right-hand side depends only on the parameters of the process followed by f_2. Define λ as the value of each side in equation (27.6), so that

$$\frac{\mu_1 - r}{\sigma_1} = \frac{\mu_2 - r}{\sigma_2} = \lambda$$

Dropping subscripts, equation (27.6) shows that if f is the price of a derivative dependent only on θ and t with

$$\frac{df}{f} = \mu\,dt + \sigma\,dz \tag{27.7}$$

then

$$\frac{\mu - r}{\sigma} = \lambda \tag{27.8}$$

The parameter λ is known as the *market price of risk* of θ. (In the context of portfolio performance measurement, it is known as the *Sharpe ratio.*) It can be dependent on both θ and t, but it is not dependent on the nature of the derivative f. Our analysis shows that, for no arbitrage, $(\mu - r)/\sigma$ must at any given time be the same for all derivatives that are dependent only on θ and t.

The market price of risk of θ measures the trade-offs between risk and return that are made for securities dependent on θ. Equation (27.8) can be written

$$\mu - r = \lambda\sigma \tag{27.9}$$

The variable σ can be loosely interpreted as the quantity of θ-risk present in f. On the right-hand side of the equation, the quantity of θ-risk is multiplied by the price of θ-risk. The left-hand side is the expected return, in excess of the risk-free interest rate, that is required to compensate for this risk. Equation (27.9) is analogous to the capital asset pricing model, which relates the expected excess return on a stock to its risk. This chapter will not be concerned with the measurement of the market price of risk. This will be discussed in Chapter 34 when the evaluation of real options is considered.

It is natural to assume that σ, the coefficient of dz, in equation (27.8) is the volatility of f. In fact, σ can be negative. This will be the case when f is negatively related the θ (so that $\partial f/\partial\theta$ is negative). It is the absolute value $|\sigma|$ of σ that is the volatility of f. One way of understanding this is to note that the process for f has the same statistical properties when we replace dz by $-dz$.

Chapter 5 distinguished between investment assets and consumption assets. An investment asset is an asset that is bought or sold purely for investment purposes by a significant number of investors. Consumption assets are held primarily for consumption.

Equation (27.8) is true for all investment assets that provide no income and depend only on θ. If the variable θ itself happens to be such an asset, then

$$\frac{m-r}{s}=\lambda$$

But, in other circumstances, this relationship is not necessarily true.

Example 27.1

Consider a derivative whose price is positively related to the price of oil and depends on no other stochastic variables. Suppose that it provides an expected return of 12% per annum and has a volatility of 20% per annum. Assume that the risk-free interest rate is 8% per annum. It follows that the market price of risk of oil is

$$\frac{0.12-0.08}{0.2}=0.2$$

Note that oil is a consumption asset rather than an investment asset, so its market price of risk cannot be calculated from equation (27.8) by setting μ equal to the expected return from an investment in oil and σ equal to the volatility of oil prices.

Example 27.2

Consider two securities, both of which are positively dependent on the 90-day interest rate. Suppose that the first one has an expected return of 3% per annum and a volatility of 20% per annum, and the second one has a volatility of 30% per annum. Assume that the instantaneous risk-free rate of interest is 6% per annum. The market price of interest rate risk is, using the expected return and volatility for the first security,

$$\frac{0.03-0.06}{0.2}=-0.15$$

From a rearrangement of equation (27.9), the expected return from the second security is, therefore,

$$0.06-0.15\times 0.3=0.015$$

or 1.5% per annum.

Alternative Worlds

The process followed by derivative price f is

$$df=\mu f\,dt+\sigma f\,dz$$

The value of μ depends on the risk preferences of investors. In a world where the market price of risk is zero, λ equals zero. From equation (27.9) $\mu=r$, so that the process followed by f is

$$df=rf\,dt+\sigma f\,dz$$

We will refer to this as the *traditional risk-neutral world.*

Other assumptions about the market price of risk, λ, enable other worlds that are internally consistent to be defined. From equation (27.9),

$$\mu=r+\lambda\sigma$$

so that

$$df = (r + \lambda\sigma) f\, dt + \sigma f\, dz \tag{27.10}$$

The market price of risk of a variable determines the growth rates of all securities dependent on the variable. As we move from one market price of risk to another, the expected growth rates of security prices change, but their volatilities remain the same. This is a general property of variables following diffusion processes and was illustrated in Section 12.7. Choosing a particular market price of risk is also referred to as defining the *probability measure*. Some value of the market price of risk corresponds to the "real world" and the growth rates of security prices that are observed in practice.

27.2 SEVERAL STATE VARIABLES

Suppose that n variables, $\theta_1, \theta_2, \dots, \theta_n$, follow stochastic processes of the form

$$d\theta_i/\theta_i = m_i\, dt + s_i\, dz_i \tag{27.11}$$

for $i = 1, 2, \dots, n$, where the dz_i are Wiener processes. The parameters m_i and s_i are expected growth rates and volatilities and may be functions of the θ_i and time. Equation (13A.10) in the appendix to Chapter 13 provides a version of Itô's lemma that covers functions of several variables. It shows that the process for the price f of a security that is dependent on the θ_i has n stochastic components. It can be written

$$df/f = \mu\, dt + \sum_{i=1}^{n} \sigma_i\, dz_i \tag{27.12}$$

In this equation, μ is the expected return from the security and $\sigma_i\, dz_i$ is the component of the risk of this return attributable to θ_i. Both μ and the σ_i are potentially dependent on the θ_i and time.

Technical Note 30 at www.rotman.utoronto.ca/~hull/TechnicalNotes shows that

$$\mu - r = \sum_{i=1}^{n} \lambda_i \sigma_i \tag{27.13}$$

where λ_i is the market price of risk for θ_i. This equation relates the expected excess return that investors require on the security to the λ_i and σ_i. Equation (27.9) is the particular case of this equation when $n = 1$. The term $\lambda_i\sigma_i$ on the right-hand side measures the extent that the excess return required by investors on a security is affected by the dependence of the security on θ_i. If $\lambda_i\sigma_i = 0$, there is no effect; if $\lambda_i\sigma_i > 0$, investors require a higher return to compensate them for the risk arising from θ_i; if $\lambda_i\sigma_i < 0$, the dependence of the security on θ_i causes investors to require a lower return than would otherwise be the case. The $\lambda_i\sigma_i < 0$ situation occurs when the variable has the effect of reducing rather than increasing the risks in the portfolio of a typical investor.

Example 27.3

A stock price depends on three underlying variables: the price of oil, the price of gold, and the performance of a stock index. Suppose that the market prices of risk for these variables are 0.2, –0.1, and 0.4, respectively. Suppose also that the σ_i in equation (27.12) corresponding to the three variables have been estimated as 0.05, 0.1, and 0.15, respectively. The excess return on the stock over the risk-free rate is

$$0.2 \times 0.05 - 0.1 \times 0.1 + 0.4 \times 0.15 = 0.06$$

or 6.0% per annum. If variables other than those considered affect the stock price, this result is still true provided that the market price of risk for each of these other variables is zero.

Equation (27.13) is closely related to arbitrage pricing theory, developed by Stephen Ross in 1976.[2] The continuous-time version of the capital asset pricing model (CAPM) can be regarded as a particular case of the equation. CAPM argues that an investor requires excess returns to compensate for any risk that is correlated to the risk in the return from the stock market, but requires no excess return for other risks. Risks that are correlated with the return from the stock market are referred to as *systematic*; other risks are referred to as *nonsystematic*. If CAPM is true, then λ_i is proportional to the correlation between changes in θ_i and the return from the market. When θ_i is uncorrelated with the return from the market, λ_i is zero.

27.3 MARTINGALES

A *martingale* is a zero-drift stochastic process.[3] A variable θ follows a martingale if its process has the form

$$d\theta = \sigma\, dz$$

where dz is a Wiener process. The variable σ may itself be stochastic. It can depend on θ and other stochastic variables. A martingale has the convenient property that its expected value at any future time is equal to its value today. This means that

$$E(\theta_T) = \theta_0$$

where θ_0 and θ_T denote the values of θ at times zero and T, respectively. To understand this result, note that over a very small time interval the change in θ is normally distributed with zero mean. The expected change in θ over any very small time interval is therefore zero. The change in θ between time 0 and time T is the sum of its changes over many small time intervals. It follows that the expected change in θ between time 0 and time T must also be zero.

The Equivalent Martingale Measure Result

Suppose that f and g are the prices of traded securities dependent on a single source of uncertainty. Assume that the securities provide no income during the time period under consideration and define $\phi = f/g$.[4] The variable ϕ is the relative price of f with respect to g. It can be thought of as measuring the price of f in units of g rather than dollars. The security price g is referred to as the *numeraire*.

[2] See S.A. Ross, "The Arbitrage Theory of Capital Asset Pricing," *Journal of Economic Theory*, 13 (December 1976): 343–62.

[3] More formally, a sequence of random variables $X_0, X_1, \ldots$ is a martingale if $E(X_i \mid X_{i-1}, X_{i-2}, , X_0) = X_{i-1}$, for all $i > 0$, where E denotes expectation.

[4] Problem 27.8 extends the analysis to situations where the securities provide income.

The *equivalent martingale measure* result shows that, when there are no arbitrage opportunities, ϕ is a martingale for some choice of the market price of risk. What is more, for a given numeraire security g, the same choice of the market price of risk makes ϕ a martingale for all securities f. This choice of the market price of risk is the volatility of g. In other words, when the market price of risk is set equal to the volatility of g, the ratio f/g is a martingale for all security prices f. (Note that the market price of risk has the same dimension as volatility. Both are "per square root of time." The choice for the market price of risk is therefore valid.)

To prove this result, suppose that the volatilities of f and g are σ_f and σ_g. From equation (27.10), in a world where the market price of risk is σ_g,

$$df = (r + \sigma_g \sigma_f) f\, dt + \sigma_f f\, dz$$

$$dg = (r + \sigma_g^2) g\, dt + \sigma_g g\, dz$$

Using Itô's lemma gives

$$d \ln f = (r + \sigma_g \sigma_f - \sigma_f^2/2)\, dt + \sigma_f\, dz$$

$$d \ln g = (r + \sigma_g^2/2)\, dt + \sigma_g\, dz$$

so that

$$d(\ln f - \ln g) = (\sigma_g \sigma_f - \sigma_f^2/2 - \sigma_g^2/2)\, dt + (\sigma_f - \sigma_g)\, dz$$

or

$$d\left(\ln \frac{f}{g}\right) = -\frac{(\sigma_f - \sigma_g)^2}{2}\, dt + (\sigma_f - \sigma_g)\, dz$$

Itô's lemma can be used to determine the process for f/g from the process for $\ln(f/g)$:

$$d\left(\frac{f}{g}\right) = (\sigma_f - \sigma_g) \frac{f}{g}\, dz \qquad \textbf{(27.14)}$$

This shows that f/g is a martingale and proves the equivalent martingale measure result. We will refer to a world where the market price of risk is the volatility σ_g of g as a world that is *forward risk neutral* with respect to g.

Because f/g is a martingale in a world that is forward risk neutral with respect to g, it follows from the result at the beginning of this section that

$$\frac{f_0}{g_0} = E_g\left(\frac{f_T}{g_T}\right)$$

or

$$f_0 = g_0 E_g\left(\frac{f_T}{g_T}\right) \qquad \textbf{(27.15)}$$

where E_g denotes the expected value in a world that is forward risk neutral with respect to g.

27.4 ALTERNATIVE CHOICES FOR THE NUMERAIRE

We now present a number of examples of the equivalent martingale measure result. The first example shows that it is consistent with the traditional risk-neutral valuation result used in earlier chapters. The other examples prepare the way for the valuation of bond options, interest rate caps, and swap options in Chapter 28.

Money Market Account as the Numeraire

The dollar money market account is a security that is worth \$1 at time zero and earns the instantaneous risk-free rate r at any given time.[5] The variable r may be stochastic. If we set g equal to the money market account, it grows at rate r so that

$$dg = rg\,dt \tag{27.16}$$

The drift of g is stochastic, but the volatility of g is zero. It follows from the results in Section 27.3 that f/g is a martingale in a world where the market price of risk is zero. This is the world we defined earlier as the traditional risk-neutral world. From equation (27.15),

$$f_0 = g_0\hat{E}\left(\frac{f_T}{g_T}\right) \tag{27.17}$$

where $\hat{E}$ denotes expectations in the traditional risk-neutral world.

In this case, $g_0 = 1$ and

$$g_T = e^{\int_0^T r\,dt}$$

so that equation (27.17) reduces to

$$f_0 = \hat{E}\left(e^{-\int_0^T r\,dt} f_T\right) \tag{27.18}$$

or

$$f_0 = \hat{E}\left(e^{-\bar{r}T} f_T\right) \tag{27.19}$$

where $\bar{r}$ is the average value of r between time 0 and time T. This equation shows that one way of valuing an interest rate derivative is to simulate the short-term interest rate r in the traditional risk-neutral world. On each trial the expected payoff is calculated and discounted at the average value of the short rate on the sampled path.

When the short-term interest rate r is assumed to be constant, equation (27.19) reduces to

$$f_0 = e^{-rT}\hat{E}(f_T)$$

or the risk-neutral valuation relationship used in earlier chapters.

Zero-Coupon Bond Price as the Numeraire

Define $P(t, T)$ as the price at time t of a zero-coupon bond that pays off \$1 at time T. We now explore the implications of setting g equal to $P(t, T)$. Let E_T denote expectations in a world that is forward risk neutral with respect to $P(t, T)$. Because $g_T = P(T, T) = 1$ and $g_0 = P(0, T)$, equation (27.15) gives

$$f_0 = P(0, T)E_T(f_T) \tag{27.20}$$

[5] The money account is the limit as Δt approaches zero of the following security. For the first short period of time of length Δt, it is invested at the initial Δt period rate; at time Δt, it is reinvested for a further period of time Δt at the new Δt period rate; at time $2\Delta t$, it is again reinvested for a further period of time Δt at the new Δt period rate; and so on. The money market accounts in other currencies are defined analogously to the dollar money market account.

Notice the difference between equations (27.20) and (27.19). In equation (27.19), the discounting is inside the expectations operator. In equation (27.20) the discounting, as represented by the $P(0, T)$ term, is outside the expectations operator. The use of $P(t, T)$ as the numeraire therefore considerably simplifies things for a security that provides a payoff solely at time T.

Consider any variable θ that is not an interest rate.[6] A forward contract on θ with maturity T is defined as a contract that pays off $\theta_T - K$ at time T, where θ_T is the value θ at time T. Define f as the value of this forward contract. From equation (27.20),

$$f_0 = P(0, T)[E_T(\theta_T) - K]$$

The forward price, F, of θ is the value of K for which f_0 equals zero. It therefore follows that

$$P(0, T)[E_T(\theta_T) - F] = 0$$

or

$$F = E_T(\theta_T) \tag{27.21}$$

Equation (27.21) shows that the forward price of any variable (except an interest rate) is its expected future spot price in a world that is forward risk neutral with respect to $P(t, T)$. Note the difference here between forward prices and futures prices. The argument in Section 17.7 shows that the futures price of a variable is the expected future spot price in the traditional risk-neutral world.

Equation (27.20) shows that any security that provides a payoff at time T can be valued by calculating its expected payoff in a world that is forward risk neutral with respect to a bond maturing at time T and discounting at the risk-free rate for maturity T. Equation (27.21) shows that it is correct to assume that the expected value of the underlying variables equal their forward values when computing the expected payoff.

Interest Rates When Zero-Coupon Bond Price is the Numeraire

For the next result, define $R(t, T, T^*)$ as the forward interest rate as seen at time t for the period between T and T^* expressed with a compounding period of $T^* - T$. (For example, if $T^* - T = 0.5$, the interest rate is expressed with semiannual compounding; if $T^* - T = 0.25$, it is expressed with quarterly compounding; and so on.) The forward price, as seen at time t, of a zero-coupon bond lasting between times T and T^* is

$$\frac{P(t, T^*)}{P(t, T)}$$

A forward interest rate is defined differently from the forward value of most variables. A forward interest rate is the interest rate implied by the corresponding forward bond price. It follows that

$$\frac{1}{[1 + (T^* - T)R(t, T, T^*)]} = \frac{P(t, T^*)}{P(t, T)}$$

so that

$$R(t, T, T^*) = \frac{1}{T^* - T}\left[\frac{P(t, T)}{P(t, T^*)} - 1\right]$$

[6] The analysis given here does not apply to interest rates because forward contracts for interest rates are defined differently from forward contracts for other variables. A forward interest rate is the interest rate implied by the corresponding forward bond price.

or

$$R(t, T, T^*) = \frac{1}{T^* - T}\left[\frac{P(t, T) - P(t, T^*)}{P(t, T^*)}\right]$$

Setting

$$f = \frac{1}{T^* - T}[P(t, T) - P(t, T^*)]$$

and $g = P(t, T^*)$, the equivalent martingale measure result shows that $R(t, T, T^*)$ is a martingale in a world that is forward risk neutral with respect to $P(t, T^*)$. This means that

$$R(0, T, T^*) = E_{T^*}[R(T, T, T^*)] \qquad \textbf{(27.22)}$$

where E_{T^*} denotes expectations in a world that is forward risk neutral with respect to $P(t, T^*)$.

The variable $R(0, T, T^*)$ is the forward interest rate between times T and T^* as seen at time 0, whereas $R(T, T, T^*)$ is the realized interest rate between times T and T^*. Equation (27.22) therefore shows that the forward interest rate between times T and T^* equals the expected future interest rate in a world that is forward risk neutral with respect to a zero-coupon bond maturing at time T^*. This result, when combined with that in equation (27.20), will be critical to an understanding of the standard market model for interest rate caps in the next chapter.

Annuity Factor as the Numeraire

For the next application of equivalent martingale measure arguments, consider a swap starting at a future time T with payment dates at times $T_1, T_2, \dots, T_N$. Define $T_0 = T$. Assume that the principal underlying the swap is \$1. Suppose that the forward swap rate (i.e., the interest rate on the fixed side that makes the swap have a value of zero) is $s(t)$ at time t ($t \leqslant T$). The value of the fixed side of the swap is

$$s(t)A(t)$$

where

$$A(t) = \sum_{i=0}^{N-1}(T_{i+1} - T_i)P(t, T_{i+1})$$

Chapter 7 showed that, when the principal is added to the payment on the last payment date of a swap, the value of the floating side of the swap on the initiation date equals the underlying principal. It follows that if \$1 is added at time T_N, the floating side is worth \$1 at time T_0. (This is because, when the discount rate is the LIBOR/swap rate, the present value of the payments on a LIBOR floating-rate bond equals the bond's principal.) The value of \$1 received at time T_N is $P(t, T_N)$. The value of \$1 at time T_0 is $P(t, T_0)$. The value of the floating side at time t is, therefore,

$$P(t, T_0) - P(t, T_N)$$

Equating the values of the fixed and floating sides gives

$$s(t)A(t) = P(t, T_0) - P(t, T_N)$$

or

$$s(t) = \frac{P(t, T_0) - P(t, T_N)}{A(t)} \tag{27.23}$$

The equivalent martingale measure result can be applied by setting f equal to $P(t, T_0) - P(t, T_N)$ and g equal to $A(t)$. This leads to

$$s(t) = E_A[s(T)] \tag{27.24}$$

where E_A denotes expectations in a world that is forward risk neutral with respect to $A(t)$. Therefore, in a world that is forward risk neutral with respect to $A(t)$, the expected future swap rate is the current swap rate.

For any security, f, the result in equation (27.15) shows that

$$f_0 = A(0)E_A\left[\frac{f_T}{A(T)}\right] \tag{27.25}$$

This result, when combined with the result in equation (27.24), will be critical to an understanding of the standard market model for European swap options in the next chapter.

27.5 EXTENSION TO SEVERAL FACTORS

The results presented in Sections 27.3 and 27.4 can be extended to cover the situation when there are many independent factors.[7] Assume that there are n independent factors and that the processes for f and g in the traditional risk-neutral world are

$$df = rf\,dt + \sum_{i=1}^{n} \sigma_{f,i} f\,dz_i$$

and

$$dg = rg\,dt + \sum_{i=1}^{n} \sigma_{g,i} g\,dz_i$$

It follows from Section 27.2 that other internally consistent worlds can be defined by setting

$$df = \left[r + \sum_{i=1}^{n} \lambda_i \sigma_{f,i}\right] f\,dt + \sum_{i=1}^{n} \sigma_{f,i} f\,dz_i$$

and

$$dg = \left[r + \sum_{i=1}^{n} \lambda_i \sigma_{g,i}\right] g\,dt + \sum_{i=1}^{n} \sigma_{g,i} g\,dz_i$$

where the λ_i $(1 \leqslant i \leqslant n)$ are the n market prices of risk. One of these other worlds is the real world.

The definition of forward risk neutrality can be extended so that a world is forward risk neutral with respect to g, where $\lambda_i = \sigma_{g,i}$ for all i. It can be shown from Itô's lemma, using the fact that the dz_i are uncorrelated, that the process followed by f/g in

[7] The independence condition is not critical. If factors are not independent they can be orthogonalized.

this world has zero drift (see Problem 27.12). The rest of the results in the last two sections (from equation (27.15) onward) are therefore still true.

27.6 BLACK'S MODEL REVISITED

Section 17.8 explained that Black's model is a popular tool for pricing European options in terms of the forward or futures price of the underlying asset when interest rates are constant. We are now in a position to relax the constant interest rate assumption and show that Black's model can be used to price European options in terms of the forward price of the underlying asset when interest rates are stochastic.

Consider a European call option on an asset with strike price K that lasts until time T. From equation (27.20), the option's price is given by

$$c = P(0, T)E_T[\max(S_T - K, 0)] \tag{27.26}$$

where S_T is the asset price at time T and E_T denotes expectations in a world that is forward risk neutral with respect to $P(t, T)$. Define F_0 and F_T as the forward price of the asset at time 0 and time T for a contract maturing at time T. Because $S_T = F_T$,

$$c = P(0, T)E_T[\max(F_T - K, 0)]$$

Assume that F_T is lognormal in the world being considered, with the standard deviation of $\ln(F_T)$ equal to $\sigma_F\sqrt{T}$. This could be because the forward price follows a stochastic process with constant volatility σ_F. The appendix at the end of Chapter 14 shows that

$$E_T[\max(F_T - K, 0)] = E_T(F_T)N(d_1) - KN(d_2) \tag{27.27}$$

where

$$d_1 = \frac{\ln[E_T(F_T)/K] + \sigma_F^2 T/2}{\sigma_F\sqrt{T}}$$

$$d_2 = \frac{\ln[E_T(F_T)/K] - \sigma_F^2 T/2}{\sigma_F\sqrt{T}}$$

From equation (27.21), $E_T(F_T) = E_T(S_T) = F_0$. Hence,

$$c = P(0, T)[F_0 N(d_1) - KN(d_2)] \tag{27.28}$$

where

$$d_1 = \frac{\ln[F_0/K] + \sigma_F^2 T/2}{\sigma_F\sqrt{T}}$$

$$d_2 = \frac{\ln[F_0/K] - \sigma_F^2 T/2}{\sigma_F\sqrt{T}}$$

Similarly,

$$p = P(0, T)[KN(-d_2) - F_0 N(-d_1)] \tag{27.29}$$

where p is the price of a European put option on the asset with strike price K and time to maturity T. This is Black's model. It applies to both investment and consumption assets and, as we have just shown, is true when interest rates are stochastic provided that F_0 is the forward asset price. The variable σ_F can be interpreted as the (constant) volatility of the forward asset price.

27.7 OPTION TO EXCHANGE ONE ASSET FOR ANOTHER

Consider next an option to exchange an investment asset worth U for an investment asset worth V. This has already been discussed in Section 25.13. Suppose that the volatilities of U and V are σ_U and σ_V and the coefficient of correlation between them is ρ.

Assume first that the assets provide no income and choose the numeraire security g to be U. Setting $f = V$ in equation (27.15) gives

$$V_0 = U_0 E_U\left(\frac{V_T}{U_T}\right) \tag{27.30}$$

where E_U denotes expectations in a world that is forward risk neutral with respect to U.

The variable f in equation (27.15) can be set equal to the value of the option under consideration, so that $f_T = \max(V_T - U_T, 0)$. It follows that

$$f_0 = U_0\, E_U\left[\frac{\max(V_T - U_T, 0)}{U_T}\right]$$

or

$$f_0 = U_0\, E_U\left[\max\left(\frac{V_T}{U_T} - 1, 0\right)\right] \tag{27.31}$$

The volatility of V/U is $\hat{\sigma}$ (see Problem 27.14), where

$$\hat{\sigma}^2 = \sigma_U^2 + \sigma_V^2 - 2\rho\sigma_U\sigma_V$$

From the appendix at the end of Chapter 14, equation (27.31) becomes

$$f_0 = U_0\left[E_U\left(\frac{V_T}{U_T}\right)N(d_1) - N(d_2)\right]$$

where

$$d_1 = \frac{\ln(V_0/U_0) + \hat{\sigma}^2 T/2}{\hat{\sigma}\sqrt{T}} \quad \text{and} \quad d_2 = d_1 - \hat{\sigma}\sqrt{T}$$

Substituting from equation (27.30) gives

$$f_0 = V_0 N(d_1) - U_0 N(d_2) \tag{27.32}$$

This is the value of an option to exchange one asset for another when the assets provide no income.

Problem 27.8 shows that, when f and g provide income at rate q_f and q_g, equation (27.15) becomes

$$f_0 = g_0 e^{(q_f - q_g)T} E_g\left(\frac{f_T}{g_T}\right)$$

This means that equations (27.30) and (27.31) become

$$E_U\left(\frac{V_T}{U_T}\right) = e^{(q_U - q_V)T}\frac{V_0}{U_0}$$

and

$$f_0 = e^{-q_U T} U_0 E_U\left[\max\left(\frac{V_T}{U_T} - 1,\ 0\right)\right]$$

and equation (27.32) becomes

$$f_0 = e^{-q_V T} V_0 N(d_1) - e^{-q_U T} U_0 N(d_2)$$

with d_1 and d_2 being redefined as

$$d_1 = \frac{\ln(V_0/U_0) + (q_U - q_V + \hat{\sigma}^2/2)T}{\hat{\sigma}\sqrt{T}} \quad \text{and} \quad d_2 = d_1 - \hat{\sigma}\sqrt{T}$$

This is the result given in equation (25.5) for the value of an option to exchange one asset for another.

27.8 CHANGE OF NUMERAIRE

In this section, we consider the impact of a change in numeraire on the process followed by a market variable. Suppose first that the variable is the price of a traded security, f. In a world where the market price of dz_i risk is λ_i,

$$df = \left[r + \sum_{i=1}^{n} \lambda_i \sigma_{f,i}\right] f\,dt + \sum_{i=1}^{n} \sigma_{f,i} f\,dz_i$$

Similarly, when it is λ_i^*,

$$df = \left[r + \sum_{i=1}^{n} \lambda_i^* \sigma_{f,i}\right] f\,dt + \sum_{i=1}^{n} \sigma_{f,i} f\,dz_i$$

The effect of moving from the first world to the second is therefore to increase the expected growth rate of the price of any traded security f by

$$\sum_{i=1}^{n} (\lambda_i^* - \lambda_i)\sigma_{f,i}$$

Consider next a variable v that is not the price of a traded security. As shown in Technical Note 20 at www.rotman.utoronto.ca/~hull/TechnicalNotes, the expected growth rate of v responds to a change in the market price of risk in the same way as the expected growth rate of the prices of traded securities. It increases by

$$\alpha_v = \sum_{i=1}^{n} (\lambda_i^* - \lambda_i)\sigma_{v,i} \tag{27.33}$$

where $\sigma_{v,i}$ is the ith component of the volatility of v.

When we move from a numeraire of g to a numeraire of h, $\lambda_i = \sigma_{g,i}$ and $\lambda_i^* = \sigma_{h,i}$. Define $w = h/g$ and $\sigma_{w,i}$ as the ith component of the volatility of w. From Itô's lemma (see Problem 27.14),

$$\sigma_{w,i} = \sigma_{h,i} - \sigma_{g,i}$$

so that equation (27.33) becomes

$$\alpha_v = \sum_{i=1}^{n} \sigma_{w,i}\,\sigma_{v,i} \tag{27.34}$$

We will refer to w as the *numeraire ratio*. Equation (27.34) is equivalent to

$$\alpha_v = \rho\sigma_v\sigma_w \tag{27.35}$$

where σ_v is the total volatility of v, σ_w is the total volatility of w, and ρ is the instantaneous correlation between changes in v and w.[8]

This is a surprisingly simple result. The adjustment to the expected growth rate of a variable v when we change from one numeraire to another is the instantaneous covariance between the percentage change in v and the percentage change in the numeraire ratio. This result will be used when timing and quanto adjustments are considered in Chapter 29.

A particular case of the results in this section is when we move from the real world to the traditional risk-neutral world (where all the market prices of risk are zero). From equation (27.33), the growth rate of v changes by $-\sum_{i=1}^{n} \lambda_i\sigma_{vi}$. This corresponds to the result in equation (27.13) when v is the price of a traded security. We have shown that it is also true when v is not the price of a traded security. In general, the way that we move from one world to another for variables that are not the prices of traded securities are the same as for those that are.

SUMMARY

The market price of risk of a variable defines the trade-offs between risk and return for traded securities dependent on the variable. When there is one underlying variable, a derivative's excess return over the risk-free rate equals the market price of risk multiplied by the derivative's volatility. When there are many underlying variables, the excess return is the sum of the market price of risk multiplied by the volatility for each variable.

A powerful tool in the valuation of derivatives is risk-neutral valuation. This was introduced in Chapters 12 and 14. The principle of risk-neutral valuation shows that, if we assume that the world is risk neutral when valuing derivatives, we get the right answer—not just in a risk-neutral world, but in all other worlds as well. In the traditional risk-neutral world, the market price of risk of all variables is zero. This chapter has extended the principle of risk-neutral valuation. It has shown that, when

[8] To see this, note that the changes Δv and Δw in v and w in a short period of time Δt are given by

$$\Delta v = \cdots + \sum \sigma_{v,i}\, v\epsilon_i\sqrt{\Delta t}$$
$$\Delta w = \cdots + \sum \sigma_{w,i}\, w\epsilon_i\sqrt{\Delta t}$$

Since the dz_i are uncorrelated, it follows that $E(\epsilon_i\epsilon_j) = 0$ when $i \neq j$. Also, from the definition of ρ, we have

$$\rho v\sigma_v w\sigma_w = E(\Delta v\,\Delta w) - E(\Delta v)\,E(\Delta w)$$

When terms of higher order than Δt are ignored this leads to

$$\rho\sigma_v\sigma_w = \sum \sigma_{w,i}\,\sigma_{v,i}$$

interest rates are stochastic, there are many interesting and useful alternatives to the traditional risk-neutral world.

A martingale is a zero drift stochastic process. Any variable following a martingale has the simplifying property that its expected value at any future time equals its value today. The equivalent martingale measure result shows that, if g is a security price, there is a world in which the ratio f/g is a martingale for all security prices f. It turns out that, by appropriately choosing the numeraire security g, the valuation of many interest rate dependent derivatives can be simplified.

This chapter has used the equivalent martingale measure result to extend Black's model to the situation where interest rates are stochastic and to value an option to exchange one asset for another. In Chapters 28 to 32, it will be useful in valuing interest rate derivatives.

FURTHER READING

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Practice Questions (Answers in the Solutions Manual)

27.1. How is the market price of risk defined for a variable that is not the price of an investment asset?

27.2. Suppose that the market price of risk for gold is zero. If the storage costs are 1% per annum and the risk-free rate of interest is 6% per annum, what is the expected growth rate in the price of gold? Assume that gold provides no income.

27.3. Consider two securities both of which are dependent on the same market variable. The expected returns from the securities are 8% and 12%. The volatility of the first security is 15%. The instantaneous risk-free rate is 4%. What is the volatility of the second security?

27.4. An oil company is set up solely for the purpose of exploring for oil in a certain small area of Texas. Its value depends primarily on two stochastic variables: the price of oil and the quantity of proven oil reserves. Discuss whether the market price of risk for the second of these two variables is likely to be positive, negative, or zero.

27.5. Deduce the differential equation for a derivative dependent on the prices of two non-dividend-paying traded securities by forming a riskless portfolio consisting of the derivative and the two traded securities.

27.6. Suppose that an interest rate x follows the process

$$dx = a(x_0 - x)\,dt + c\sqrt{x}\,dz$$

where a, x_0, and c are positive constants. Suppose further that the market price of risk for x is λ. What is the process for x in the traditional risk-neutral world?

27.7. Prove that, when the security f provides income at rate q, equation (27.9) becomes $\mu + q - r = \lambda\sigma$. (*Hint*: Form a new security f^* that provides no income by assuming that all the income from f is reinvested in f.)

27.8. Show that when f and g provide income at rates q_f and q_g, respectively, equation (27.15) becomes

$$f_0 = g_0 e^{(q_f - q_g)T} E_g\left(\frac{f_T}{g_T}\right)$$

(*Hint*: Form new securities f^* and g^* that provide no income by assuming that all the income from f is reinvested in f and all the income in g is reinvested in g.)

27.9. "The expected future value of an interest rate in a risk-neutral world is greater than it is in the real world." What does this statement imply about the market price of risk for (a) an interest rate and (b) a bond price. Do you think the statement is likely to be true? Give reasons.

27.10. The variable S is an investment asset providing income at rate q measured in currency A. It follows the process

$$dS = \mu_S S\,dt + \sigma_S S\,dz$$

in the real world. Defining new variables as necessary, give the process followed by S, and the corresponding market price of risk, in:
(a) A world that is the traditional risk-neutral world for currency A
(b) A world that is the traditional risk-neutral world for currency B
(c) A world that is forward risk neutral with respect to a zero-coupon currency A bond maturing at time T
(d) A world that is forward risk neutral with respect to a zero coupon currency B bond maturing at time T.

27.11. Explain the difference between the way a forward interest rate is defined and the way the forward values of other variables such as stock prices, commodity prices, and exchange rates are defined.

27.12. Prove the result in Section 27.5 that when

$$df = \left[r + \sum_{i=1}^{n} \lambda_i \sigma_{f,i}\right] f\,dt + \sum_{i=1}^{n} \sigma_{f,i} f\,dz_i$$

and

$$dg = \left[r + \sum_{i=1}^{n} \lambda_i \sigma_{g,i}\right] g\,dt + \sum_{i=1}^{n} \sigma_{g,i} g\,dz_i$$

with the dz_i uncorrelated, f/g is a martingale for $\lambda_i = \sigma_{g,i}$. (*Hint*: Start by using equation (13A.11) to get the processes for $\ln f$ and $\ln g$.)

27.13. Show that when $w = h/g$ and h and g are each dependent on n Wiener processes, the ith component of the volatility of w is the ith component of the volatility of h minus the ith component of the volatility of g. (*Hint*: Start by using equation (13A.11) to get the processes for $\ln f$ and $\ln g$.)

Further Questions

27.14. A security's price is positively dependent on two variables: the price of copper and the yen/dollar exchange rate. Suppose that the market price of risk for these variables is 0.5 and 0.1, respectively. If the price of copper were held fixed, the volatility of the security would be 8% per annum; if the yen/dollar exchange rate were held fixed, the volatility of the security would be 12% per annum. The risk-free interest rate is 7% per annum. What is the expected rate of return from the security? If the two variables are uncorrelated with each other, what is the volatility of the security?

27.15. Suppose that the price of a zero-coupon bond maturing at time T follows the process

$$dP(t, T) = \mu_P P(t, T)\, dt + \sigma_P P(t, T)\, dz$$

and the price of a derivative dependent on the bond follows the process

$$df = \mu_f f\, dt + \sigma_f f\, dz$$

Assume only one source of uncertainty and that f provides no income.
(a) What is the forward price F of f for a contract maturing at time T?
(b) What is the process followed by F in a world that is forward risk neutral with respect to $P(t, T)$?
(c) What is the process followed by F in the traditional risk-neutral world?
(d) What is the process followed by f in a world that is forward risk neutral with respect to a bond maturing at time T^*, where $T^* \neq T$? Assume that σ_P^* is the volatility of this bond.

27.16. Consider a variable that is not an interest rate:
(a) In what world is the futures price of the variable a martingale?
(b) In what world is the forward price of the variable a martingale?
(c) Defining variables as necessary, derive an expression for the difference between the drift of the futures price and the drift of the forward price in the traditional risk-neutral world.
(d) Show that your result is consistent with the points made in Section 5.8 about the circumstances when the futures price is above the forward price.